

WINDOWS POWERSHELL LAB REPORT



BECOMING A POWER USERTEMS, AND PACKAGE INSTALLATION

Exploring User Management,
File Systems, and Package
Installation

ABSTRACT

Dive into the essentials of operating systems—manage users, control permissions, and install software like a pro. Build practical skills that make you a confident and capable system user.

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Introduction

This report documents practical exercises using Windows PowerShell to manage users, groups, and files, as well as install and verify software using the command line. The tasks reinforce key concepts in system administration, permissions, and automation.

PART 1: SYSTEM SETUP: USER, GROUP, AND FOLDER CONFIGURATION

User Account Management

Goal: Establish a dedicated user account for managing technical tasks or toolkit operations, ensuring separation of roles and access control.

Commands Used: `New-LocalUser -Name "TechUser" -Password (ConvertTo-SecureString "P@ssw0rd123" -AsPlainText -Force) -FullName "Tech User" -Description "Toolkit User"`

Screenshot



```
Administrator: Windows PowerShell
Windows PowerShell
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Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Windows\system32> New-LocalUser -Name "TechUser" -Password (ConvertTo-SecureString "P@ssw0rd123" -AsPlainText -Force) -FullName "Tech User" -Description "Toolkit User"
>>
>>

Name      Enabled Description
----      -
TechUser  True    Toolkit User

PS C:\Windows\system32>
```

Added the user to the "Users" group.

Goal: Assign the new user to the default permission group, granting standard access rights while maintaining system security.

Commands Used: Add-LocalGroupMember -Group "Users" -Member "TechUser"

Screenshot

```
PS C:\Windows\system32> Add-LocalGroupMember -Group "Users" -Member "TechUser"
>>
>>
PS C:\Windows\system32>
```

Created a folder named C:\Tech

Goal: Set up a workspace directory for the new user to store files, configurations, or tools related to their role.

Commands Used: New-Item -Path "C:\Tech" -ItemType Directory

Screenshot

```
PS C:\Windows\system32> New-Item -Path "C:\Tech" -ItemType Directory
>>
>>

Directory: C:\

Mode                LastWriteTime         Length Name
----                -
d-----         10/16/2025  10:09 AM              Tech

PS C:\Windows\system32>
```

Granted full control to the user

Goal: Ensure the user has complete access to read, write, and modify contents within their designated workspace, enabling productivity without restrictions.

Command Used: `icacls "C:\Tech" /grant "TechUser:(OI)(CI)F"`

Screenshot

```
PS C:\Windows\system32> icacls "C:\Tech" /grant "TechUser:(OI)(CI)F"
>>
>>
processed file: C:\Tech
Successfully processed 1 files; Failed processing 0 files
PS C:\Windows\system32>
```

Verify permissions.

Goal: Confirm that the correct access rights were applied, validating the setup and preventing future permission issues.

Command: `icacls "C:\Tech"`

Screenshot

```
PS C:\Windows\system32> icacls "C:\Tech"
>>
>>
C:\Tech DESKTOP-03ME4CP\TechUser:(OI)(CI)(F)
        BUILTIN\Administrators:(I)(OI)(CI)(F)
        NT AUTHORITY\SYSTEM:(I)(OI)(CI)(F)
        BUILTIN\Users:(I)(OI)(CI)(RX)
        NT AUTHORITY\Authenticated Users:(I)(M)
        NT AUTHORITY\Authenticated Users:(I)(OI)(CI)(IO)(M)

Successfully processed 1 files; Failed processing 0 files
PS C:\Windows\system32>
```

Explanation

I used PowerShell to create a local user named TechUser and added them to the default Users group. I then created a folder C:\Tech and granted full access permissions to the user using icacls. This ensures the user can read, write, and modify files in that directory.

✓ Part 2: Install a Software Package Using CLI

Installed Chocolatey (Windows package manager)

Goal: Enable command-line-based software installation and management, streamlining deployment and updates for future tools.

Command used: Set-ExecutionPolicy Bypass -Scope Process -Force; `[System.Net.ServicePointManager]::SecurityProtocol = [System.Net.ServicePointManager]::SecurityProtocol -bor 3072; ` iex ((New-Object System.Net.WebClient).DownloadString('https://community.chocolatey.org/install.ps1'))

Screenshot

```
PS C:\Windows\system32> Set-ExecutionPolicy Bypass -Scope Process -Force; `[System.Net.ServicePointManager]::SecurityProtocol = [System.Net.ServicePointManager]::SecurityProtocol -bor 3072; ` iex ((New-Object System.Net.WebClient).DownloadString('https://community.chocolatey.org/install.ps1'))
>>
>>
WARNING: 'choco' was found at 'C:\ProgramData\chocolatey\bin\choco.exe'.
WARNING: An existing Chocolatey installation was detected. Installation will not continue. This script will not overwrite existing installations.
If there is no Chocolatey installation at 'C:\ProgramData\chocolatey', delete the folder and attempt the installation again.

Please use choco upgrade chocolatey to handle upgrades of Chocolatey itself.
If the existing installation is not functional or a prior installation did not complete, follow these steps:
- Backup the files at the path listed above so you can restore your previous installation if needed.
- Remove the existing installation manually.
- Rerun this installation script.
- Reinstall any packages previously installed, if needed (refer to the lib folder in the backup).

Once installation is completed, the backup folder is no longer needed and can be deleted.
PS C:\Windows\system32>
```

Installed Notepad++ using Chocolatey

Goal: Enable command-line-based software installation and management, streamlining deployment and updates for future tools.

Command used: `choco install notepadplusplus -y`.

Screenshot

```
PS C:\Windows\system32> choco install notepadplusplus -y
>>
Chocolatey v2.5.1
Installing the following packages:
notepadplusplus
By installing, you accept licenses for the packages.
Downloading package from source 'https://community.chocolatey.org/api/v2/'
Progress: Downloading chocolatey-compatibility.extension 1.0.0... 100%

chocolatey-compatibility.extension v1.0.0 [Approved]
chocolatey-compatibility.extension package files install completed. Performing other installation steps.
Installed/updated chocolatey-compatibility extensions.
The install of chocolatey-compatibility.extension was successful.
Deployed to 'C:\ProgramData\chocolatey\extensions\chocolatey-compatibility'
Downloading package from source 'https://community.chocolatey.org/api/v2/'
Progress: Downloading chocolatey-core.extension 1.4.0... 100%

chocolatey-core.extension v1.4.0 [Approved]
chocolatey-core.extension package files install completed. Performing other installation steps.
Installed/updated chocolatey-core extensions.
The install of chocolatey-core.extension was successful.
Deployed to 'C:\ProgramData\chocolatey\extensions\chocolatey-core'
Downloading package from source 'https://community.chocolatey.org/api/v2/'
Progress: Downloading notepadplusplus.install 8.8.6... 100%

notepadplusplus.install v8.8.6 [Approved]
notepadplusplus.install package files install completed. Performing other installation steps.
Installing 64-bit notepadplusplus.install...
notepadplusplus.install has been installed.
WARNING: No registry key found based on 'Notepad\+\+*'
notepadplusplus.install installed to 'C:\Program Files\Notepad++'
Added C:\ProgramData\chocolatey\bin\notepad++.exe shim pointed to 'c:\program files\notepad++\notepad++.exe'.
notepadplusplus.install can be automatically uninstalled.
The install of notepadplusplus.install was successful.
Software installed as 'exe', install location is likely default.
Downloading package from source 'https://community.chocolatey.org/api/v2/'
Progress: Downloading notepadplusplus 8.8.6... 100%

notepadplusplus v8.8.6 [Approved]
notepadplusplus package files install completed. Performing other installation steps.
The install of notepadplusplus was successful.
Software install location not explicitly set, it could be in package or
default install location of installer.

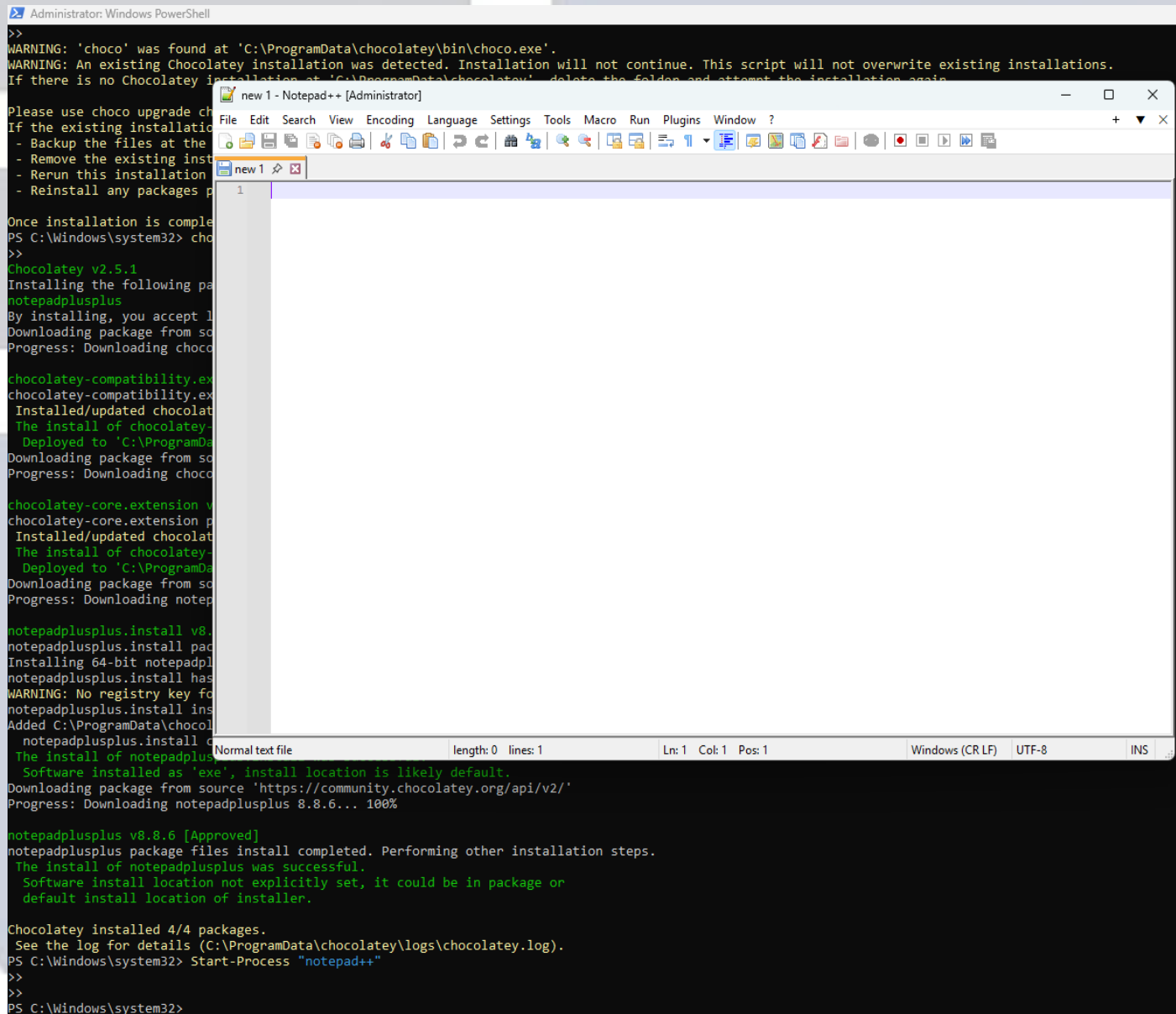
Chocolatey installed 4/4 packages.
See the log for details (C:\ProgramData\chocolatey\logs\chocolatey.log).
PS C:\Windows\system32>
```

Launched Notepad++ to verify installation

Goal: Confirm successful installation and functionality of the software, ensuring readiness for use in technical workflows.

Command used: Start-Process "notepad++"

Screenshot



The screenshot displays two overlapping windows. The background window is 'Administrator: Windows PowerShell', showing the execution of the 'choco install notepadplusplus' command. The output indicates that Notepad++ v8.8.6 was successfully installed. The foreground window is 'new 1 - Notepad++ [Administrator]', which is an empty Notepad++ editor window, confirming the application is installed and running.

```
>>>
WARNING: 'choco' was found at 'C:\ProgramData\chocolatey\bin\choco.exe'.
WARNING: An existing Chocolatey installation was detected. Installation will not continue. This script will not overwrite existing installations.
If there is no Chocolatey installation at 'C:\ProgramData\chocolatey', delete the folder and attempt the installation again.

Please use choco upgrade choco
If the existing installation is corrupt, you can try:
- Backup the files at the location
- Remove the existing installation
- Rerun this installation
- Reinstall any packages previously installed

Once installation is complete, you can run:
PS C:\Windows\system32> choco install notepadplusplus
>>>
Chocolatey v2.5.1
Installing the following packages:
notepadplusplus
By installing, you accept the license agreement at: https://notepad-plus-plus.org/about
Downloading package from source 'https://community.chocolatey.org/api/v2/'
Progress: Downloading chocolatey-core.extension v1.4.0
chocolatey-core.extension v1.4.0
chocolatey-core.extension package files install completed. Performing other installation steps.
The install of chocolatey-core.extension was successful.
Deployed to 'C:\ProgramData\chocolatey\bin'
Downloading package from source 'https://community.chocolatey.org/api/v2/'
Progress: Downloading notepadplusplus.install v8.8.6
notepadplusplus.install v8.8.6
notepadplusplus.install package files install completed. Performing other installation steps.
Installing 64-bit notepadplusplus
notepadplusplus.install has been installed.
WARNING: No registry key found for notepadplusplus.install
Added C:\ProgramData\chocolatey\bin\notepadplusplus.install.v8.8.6 to the path
The install of notepadplusplus was successful.
Software installed as 'exe', install location is likely default.
Downloading package from source 'https://community.chocolatey.org/api/v2/'
Progress: Downloading notepadplusplus 8.8.6... 100%

notepadplusplus v8.8.6 [Approved]
notepadplusplus package files install completed. Performing other installation steps.
The install of notepadplusplus was successful.
Software install location not explicitly set, it could be in package or default install location of installer.

Chocolatey installed 4/4 packages.
See the log for details (C:\ProgramData\chocolatey\logs\chocolatey.log).
PS C:\Windows\system32> Start-Process "notepad++"
>>>
>>>
PS C:\Windows\system32>
```


Explanation

I installed Chocolatey using PowerShell and then used it to install Notepad++, a popular text editor. This demonstrates how CLI-based package managers simplify software deployment. I verified the installation by launching the application directly from the terminal.

Conclusion & Reflection

This assignment gave me a practical foundation in Windows system administration using PowerShell. By creating a user (), managing permissions for the folder, and installing Notepad++ via Chocolatey, I applied key concepts in user management, file access control, and command-line software deployment.

Each task reinforced how powerful and efficient CLI tools can be for real-world administrative work. I now feel more confident using PowerShell to automate and manage system tasks, and I understand how these skills contribute to secure, scalable IT environments.

What I Learned

- PowerShell is a versatile tool for managing users, files, and software.
- Permission control is essential for protecting data and defining user roles.
- Package managers like Chocolatey simplify software installation and updates.

This project has prepared me to take on more advanced scripting and automation challenges, and I am excited to keep building tools that help teams work smarter.

. Objective

This task develops hands-on experience in Windows system administration, including user management, file permissions, and software installation via the command line.

. Practical Implementation Summary

Steps included creating a user, managing file permissions, and installing a software package using PowerShell and Chocolatey.

Conclusion & Reflection of What I learnt

This assignment provided hands-on experience in system administration using PowerShell. By creating a user account, managing permissions, and installing software using Chocolatey, I demonstrated the power and efficiency of command-line tools in managing Windows environments.

Through this exercise, I learned how permissions control enhances security, and how CLI automation simplifies administrative tasks. Going forward, I aim to explore advanced PowerShell scripting to automate larger workflows.